Motion in One Dimension

- Position x is measured from a chosen origin.
- Velocity is the rate of change of position:
  - average velocity uses two times,
  - instantaneous velocity uses a limit.
- Acceleration is the rate of change of velocity.

Constant Acceleration

When the acceleration a is constant,

$$v = v_0 + at,$$
$$x = x_0 + v_0 t + \tfrac{1}{2} a t^2.$$

Eliminating the time gives

$$v^2 = v_0^2 + 2a\left(x - x_0\right).$$

Velocity Graph

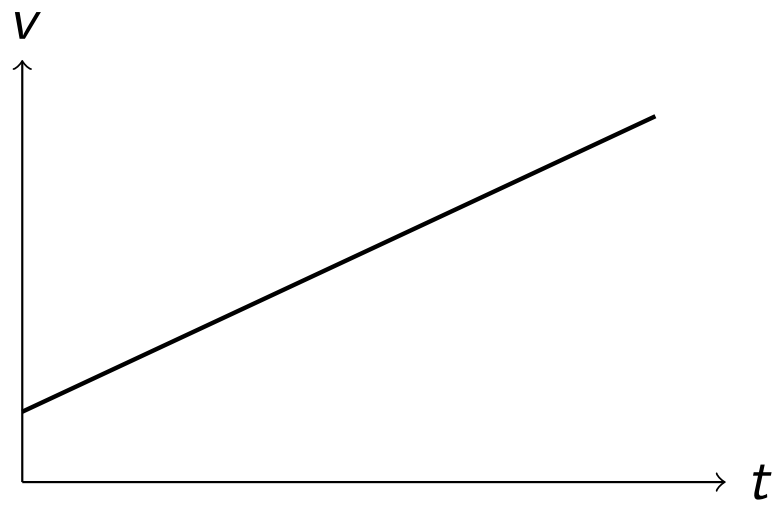

Figure: Velocity rises linearly when the acceleration is constant.